

Lecture 39

4/29/26
JRA

Housekeeping

See email w/ information re final exam

- In-Class component - very similar to Practice Final
- Take-Home component - image processing questions

MRI Contrast Review

Helpful
Weighting
Table

	TR		TE	
T_1	Short	400-600 ms	Short	5-30ms
T_2	Long	2000-4000 ms	Long	60-150ms
ρ	Long	2000-4000 ms	Short	5-30 ms

The doctor can adjust two parameters to manipulate contrast.

TR repetition time
TE echo time
= signal collection time

Other parameters - T_1 , T_2 , ρ - are intrinsic to the sample (i.e. how different parts of sample are distinguished)

T_1 weighting

Short repetition time

Short echo time

T_2 weighting

Long repetition time

Long echo time

ρ weighting

Long repetition time

Short echo time

Digital Image Processing (continued)

Continue today w/ digital image processing, focusing on image "restoration"

Images can be manipulated in both the spatial domain & frequency domain

Restoration Scheme

<u>Spatial Domain</u>	<u>Frequency Domain</u>
• Flat Field Correction * Brightness & Contrast Restoration * γ Correction * Noise Reduction • Fine Detail Restoration	* Fourier Filtering (e.g., MRI)

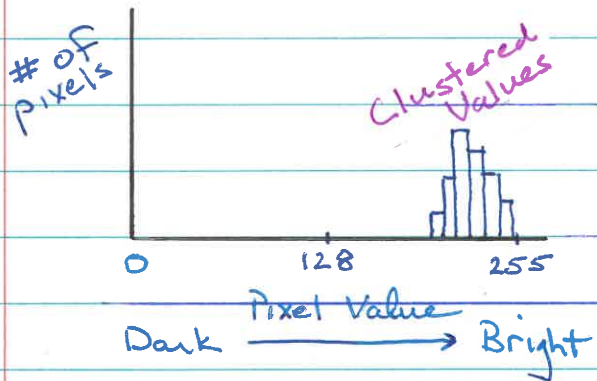
We'll look at the starred (*) techniques

* Brightness & Contrast Restoration

There are two commonly used methods for improving contrast

Do examples w/ 8-bit 256-level image

(1) Histogram Image



Values in this example are clustered together, all with high values, so the entire image will be bright

The histogram helps define the problem (e.g., all values clustered on the bright end), although the problem may be clear from the appearance of the image

The goal is a histogram that uses all grayscale levels, corresponding to an image that ranges from dark to bright

(2) Look-Up Table (LUT)

A look-up table is a rule that maps an input pixel value into an output pixel value

The look-up table can be used to create a new spread-out histogram

Histogram Stretching to enhance contrast

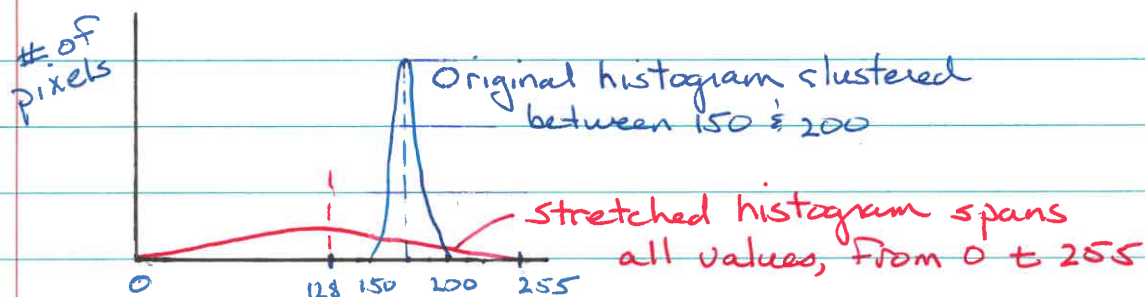
ANIMATION Mol Expressions Histogram Equalization

Looked a Fava Bean image, before & after stretching, & the associated grayscale histograms

The original image & histogram were biased toward dim, low values

The stretched image & histogram were spread out from dim, low values to bright, high values

Example



- Map ^{Input} 150 into ^{Output} 0 Totally black
- Map 200 into 255 Totally white

Do a linear interpolation in between (e.g., 175 (the center of the original distribution) maps into 128 (the center of the stretched distribution))

This is all done by a computer

*

Gamma (γ) Correction

to enhance dimmer features
 or to enhance brighter features

Gamma (γ) correction is a nonlinear tool to adjust the relative brightnesses of dimmer & brighter pixels, enhancing certain pixels

This is often used in light-stained techniques to bring up dimmer features

Example

Looked at how to use Fiji image processing software to enhance dim features in an image. Bethe took of dense core granules

Fiji Process > Math > γ

This technique can be especially helpful w/ live imaging, where you don't have much light

$$\text{Output} = 255 \left(\frac{\text{Input}}{255} \right)^\gamma$$

Exponential (nonlinear) scaling

$\gamma < 1$ Enhances dimmer pixels
 $\gamma > 1$ Enhances brighter pixels

This is a nonlinear look-up-table, in contrast to the linear table used w/ histogram stretching

*

Noise Reduction

to reduce "noise" (random, uninteresting variations) in an image

Noise degrades image quality & can interfere w/ both aesthetics & interpretation

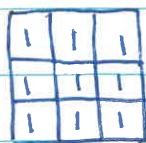
Noise is a particular problem in nuclear imaging, due to the low signal ✓

There are two popular filters for removing noise from digital images

- Mean Filter
- Median Filter

Both filters replace a pixel value in an image by a "mean" or a "median" value constructed using values of the pixel itself & its neighbors

The neighbors used to construct the new value are selected by a "kernel" & can be a group of 3×3 neighbors or 5×5 neighbors, among others



3×3 Kernel

The kernel is a matrix of values (here, all "1")

• Mean Filter

The mean filter reduces noise by replacing each pixel value by a "mean", or average, of pixel values in a pertinent part of the image (as determined by the kernel)

ANIMATION

Mol Expressions

Looked at use of 3×3 Kernel to "smooth" an image with a mean filter

48	1	22	79	-
7	54	57	39	-
72	98	33	82	-
-	-	-	-	-
-	-	-	-	-

Original Image

1	1	1
1	1	1
1	1	1

Kernel

x	x	x	x	x
x	44	52		x
x				x
x				x
x	x	x	x	x

Smoothed Image

The kernel is used to select the image pixels to be used in the smoothing operation & to determine how each is weighted

Here, applied the 3×3 kernel to the top left 3×3 set of image values.

Multiply the kernel values by the respective image values, add the resulting 9 numbers, divide by 9 to get the mean, & replace the central pixel value by the mean.

$$\boxed{} = (48.1 + 1.1 + 22.1 + 7.1 + 54.1 + 57.1 + 72.1 + 98.1 + 33.1) / 9$$

$$= 392 / 9 = 43.\bar{5}$$

$$\boxed{1} = 44$$

Repeat the process for every interior pixel, using the original (NOT smoothed) values for the calculation

$$\boxed{} = (1.1 + 22.1 + 49.1 + 54.1 + 39.1 + 98.1 + 33.1 + 82.1) / 9$$

$$= 465 / 9 = 51.\bar{6}$$

$$\boxed{} = 52$$

Bethe repeated the process, running the simulation until it populated the entire image interior w/ smoothed values.

The resulting smoothed image had much less dynamic range than the original (i.e., the pixel values were much more tightly clustered).

Image has less noise but is much more uniform.

Smoothed images have poorer resolution.

Reduce noise BUT also reduce resolution.

• Median Filter

The median filter reduces noise by replacing each pixel value by the "median" value among the pixels sampled by the kernel.

The median value is the middlemost actual value, where half of the values are lower, i.e. half of the values are higher than the median.

Apply median filter to image processed by the mean filter

48 1 7 22 33 48 54 57 72 98

54 1 22 33 39 54 57 79 82 98

The median is much less susceptible to outliers, because outliers just end up to one extreme or the other of the values below or above the median.

The median Filter removes noise but w/ much less smoothing than the mean filter.

A downside of the median Filter is that it is computationally more intensive due to the need to sort.

Example

Looked at images of raven from text

- Original - salt & pepper noise
- Mean Filter - less noise but noticeable loss of fine detail (it looks blurry)

Median Filter - less noise & less blurring
 $\Rightarrow$ the better choice

These filters cause image "shrinkage" because they aren't used to smooth edge values. However, real images are quite large, so slight shrinkage unimportant. Alternatively, other methods could be used w/ edges.

* Fourier Filter

The previous techniques were used to process (e.g., restore) images in the spatial domain. They acted on actual values of the image.

The Fourier filter, in contrast, alters the image by manipulating its frequency components in the "frequency domain".

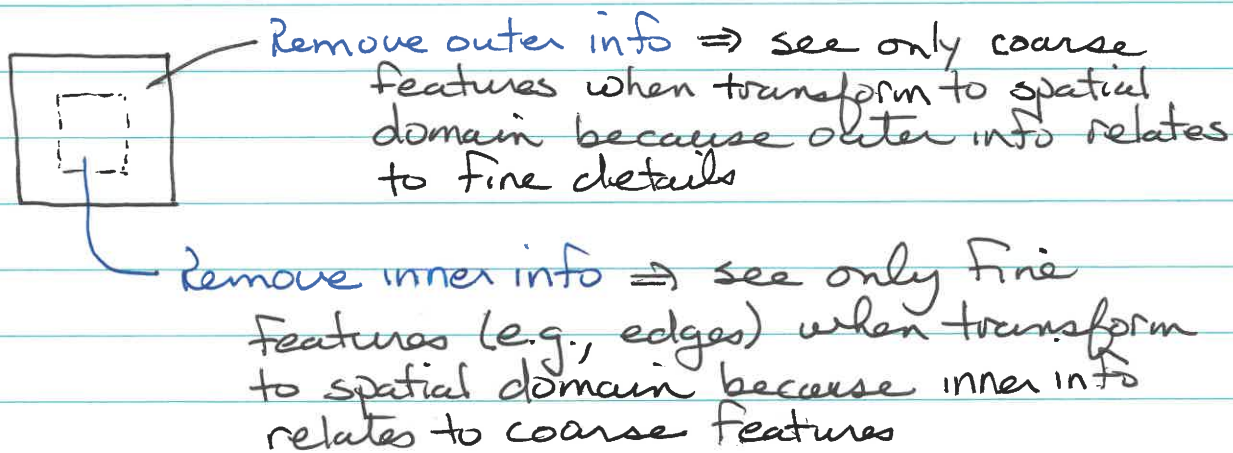
Standard images can be Fourier transformed into the frequency domain to allow Fourier filtering.

MRI images are collected in the frequency domain & can be processed before they are converted into spatial domain images.

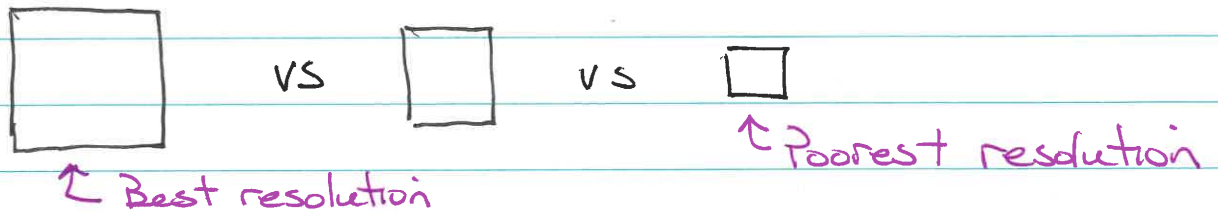
Example 1 - NASA image w/ lots of periodic noise (stripes)

- Fourier transformed using Fiji image processing software
- Removed bright periodic spots (diffraction pattern) corresponding to periodic noise
- Back transformed to return to the spatial domain & saw markedly cleaned up image

Example 2 - MRI image of brain (in frequency space)



Amount of frequency-domain data collected governs resolution of MRI image



Closing remarks ... best wishes ... Thank You!